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T13-67
SEPM Assignment 2

Q1 Explain System Development Lifecycle in detail.

→ System development Lifecycle is a structured process used to develop high quality software in a systematic and efficient way. It defines different stages involved in software development from planning to maintenance.

Phases of SDLC :- 1) Planning :- identify project & scope.
- Analyze feasibility (technical, financial, operational)
- Create project plan.

2. Requirement Analysis :- • Gather user requirements
- prepare software requirement specification (SRS)
- Define system functionalities.

3. System Design :- • Convert requirements into design
- Includes: • High-level Design (architecture)
- Low-level Design (modules, database)

4. Implementation (coding) :- • Developers write code based on design.
- Each module is developed separately.

5 Testing :- • Identify and fix errors
- Types : • Unit testing
  - Integration testing
  - System testing.

6 Deployment :- • Software is delivered to users
• Installed in real environments

7. Maintenance :- • Fix bugs and update System
• Improve performance over time

SDLC ensures systematic development, better quality and reduced risks in software projects.

Q2 Define a project? What is project management? Explain various characteristics of project management.

→ A project is a temporary activity undertaken to create a unique product, service or result.

Project management is the process of planning, organizing and controlling resources to achieve project objectives within time, cost & scope.

Characteristics of a project
1. Temporary Nature
• Has a definite Start & end
2. Unique output
• produces something new
3. Defined Objectives
• Specific goals must be achieved.
4. Limited Resources
• Uses time, money, manpower.
5. Interdependent Tasks.
• Activities depend on each other
6. Risk and uncertainty

- may face unexpected challenges.

Projects 4 goal-oriented and require proper management to ensure success.

Q3 Write a short note on PMBOK

→ PMBOK is a globally recognized guide that provides standard practices for project management.

Key features :- • Developed by Project Management Institute.
- Contains guidelines, best practices and processes
- Applicable to all types of projects.

Knowledge Areas (Examples) :- • Scope management.
- Time Management
- Cost management
- Quality Management
- Risk management.

Process Groups

1) ~~Initi~~ Initiation
2) Planning
3) Execution
4) Monitoring & controlling
5) Closing.

PMBOK helps project managers follow a structured

approach to sucessfully complete projects.

Q4 Explain ITPM

→ ITPM is the process of planning, organizing, executing and controlling IT related projects to achieve specific objectives within constraints like time, cost and scope.
It focuses specifically on managing projects related to Software development, networking, databases and IT systems.

Key components of ITPM

1. Project planning :- • Define Scope, objectives and deliverables
• prepare schedule and budget
• Allocate resources

2. Execution :- • Develop Software or implement IT systems
• Coordinate team activities.
• Ensure work is done as per plan.

3. Monitoring and control :- • Track project progress
• Compare actual vs planned performance
• Take corrective actions if needed.

4. Risk Management :- • Identify possible risks (technical, financial, operational)
• Develop strategies to minimize risks.

5 Quality Management :- • Ensure Software/System meets required standards.

- Conduct testing and reviews

Tools and Techniques Used
- Gantt charts
- Work Breakdow Structure (WBS)
- Scheduling tools
- Project management software

Importance of ITPM
- Improves efficiency of IT projects
- Reduces project failure chances
- Ensures proper resource utilization
- Helps in delivering reliable IT solutions.

Or What is role of RFP? Describe its content & purpose?

→ A Request for proposal is a formal document issued by an organization to invite vendors or suppliers to submit proposals for providing products or services for a specific project. It is widely used in project management & procurement processes.

Role of RFP :- The RFP plays an important role in selecting the most suitable vendor and ensuring smooth project execution.

1. Vendor selection:- • Helps organizations compare different vendors and choose the best one based on requirements.
2. Clear communication:- • Clearly defines project needs, expectations and scope to avoid misunderstandings.

3. Standardized process: • provides a structured format for all vendors to submit proposals ensuring fairness

4. Risk Reduction :- • Reduces chances of project failure by selecting capable and qualified vendors

5. Cost & quality control:- • Helps evaluate proposals based on ~~and~~ budget and quality, ensuring value for money.

Contents of an RFP :- An RFP generally includes the following components:

1. Project overview:- • Brief description of the Organization & project background.

2. Scope of work:- • Detailed explanation of tasks, deliverables and responsibilities.

3. Requirements :- • Technical, functional and performance requirements of the project.

4. Timeline: • Project schedule, milestones & deadlines.

5. Budget Information:- • Estimated cost or budget constraints

6. Evaluation Criteria:- • Criteria used to assess vendor proposals (cost, experiences, quality etc).

7. Terms and Conditions :- legal rules, payment terms and contract conditions.

8. Submission Guidelines:- • Instructions for proposal format, deadline and submission method

Purpose of RFP:- • To invite competitive proposals from vendors.

• To ensure transperency & fairness in selection.

- To clearly define project expectations.
- To select the most efficient and cost-effective solution.

Q6 What is feasibility Study? Explain its types, contents and purpose.

:) A feasibility study is an analysis conducted to determine whether a project is viable and worth implementing.

Types of feasibility :- 1) Technical feasibility :- Availability of technology & skills.
2 Economic feasibility :- cost vs benefits analysis.
3 Operational feasibility :- System usability and acceptance
4 Legal feasibility :- compliance with laws.
5 Schedule feasibility :- can the project be completed on time.

Contents
- problem Statement
- proposed solution
- cost estimation
- Risk analysis
- feasibility results

Purpose
- Reduce risk
- Improve decision making
- Avoid project failure.

Feasibility study ensures that only practical & beneficial

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projects are selected.

Q2 Write a short note on W.B.S

→ WBS is a hierarchical breakdown of a project into smaller manageable tasks.

Structure

Project
- Phase 1
  - Task 1
  - Task 2
- Phase 2
  - Task 3
  - Task 4

Key features :- Divides project into smaller units
- Helps in planning and scheduling
- Assigns responsibilities

Benefits :- • Better project control
- easy monitoring
- Accurate cost estimation
- Improved teamwork.

WBS simplifies complex projects and ensures efficient execution.